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How multiplayer online games can yield positive effects on individual gamers, gaming companies, and society as a whole

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Besides being a robust platform for communication and interaction among gamers, multiplayer online gaming could foster a win–win–win scenario for gamers, gaming companies, and the global society. This study developed a comprehensive model that meticulously delineates players' behaviors and cognitive patterns within multiplayer online gaming settings. This model also incorporates social interactions in gaming and in gaming communities, self-efficacy in gaming, self-awareness, brand intimacy, and a sense of social connection among remote players. Data from a total of 844 surveys were extracted from two distinct player cohorts—one engaged in charitable activities and the other not. Model validation was performed via structural equation modeling using smartPLS, which indicated that philanthropic endeavors can amplify players' brand intimacy and feelings of social connection with distant players. However, philanthropic activities can also positively or negatively moderate the relation between certain variables. The insights from this study can equip gaming companies with strategic knowledge to benefit players and the global community as they promote their business objectives.

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Introduction

Multiplayer online games allow players to engage in global communication and social interaction. These platforms incorporate multilingual features to bridge diverse cultures, fostering virtual camaraderie that may help address global challenges (Adger 2010). Initiatives such as Humble Bundle are already advancing this cause, demonstrating how the power of gaming can drive positive change (bcrf.org). However, the role played by gaming in building social cohesion remains underresearched (Kowert et al. 2014). Beyond connecting players, stakeholders must also develop strategies that mutually benefit gaming companies, players, and the global community—an area that deserves greater focus.

This study aims to foster a win–win–win scenario for gaming companies, players, and the global community using the potential of multiplayer online games. For gaming companies, this study draws on the concept of brand intimacy, which has provided significant value to businesses (Park et al. 2006; Keller and Aaker 1992) while also building meaningful connections among their customers (Almubarak et al. 2018). For players, we propose the cultivation of private self-awareness, a critical factor in gamers' personal growth (Ugur et al. 2015).

To develop a model for this investigation, this study went outside the frequently examined aspects of social interaction in gaming and in gaming communities (Merhi 2016; Jung 2020). Instead, it analyzed players' online behaviors and, building on the literature, identified another variable essential to this goal: self-efficacy in gaming. Research has indicated that players often develop self-efficacy within gaming contexts (Wu and Wang 2024). Strengthening this trait allows players to approach challenges with higher confidence, fostering both their personal growth and development (Bandura 1995).

Recently, the importance of charity events hosted by gaming brands has become increasingly evident, as demonstrated by Xbox's partnership with the Special Olympics (news.xbox.com) and Nintendo's support for various causes (www.nintendo.co.jp). By actively initiating player participation in philanthropic initiatives, gaming brands go further than mere entertainment and initiate psychological change among gamers. However, the potential psychological impact of gaming on philanthropy has been underexplored, and understanding how game companies can adapt their strategies would contribute highly to this emerging field.

Accordingly, to develop the comprehensive model with a focus on multiplayer online games, this study considered social interaction within gaming as its foundational factor, influencing both self-efficacy in gaming and interactions within gaming communities. Furthermore, we postulated that self-efficacy in gaming can significantly affect social interactions within gaming circles.

Next, we proposed the core of this model, which links multiplayer online gaming variables—self-efficacy in gaming, social interactions in gaming, and social interactions in gaming communities—with three benefit factors: feelings of social connection with distant other players (FSDOP), brand intimacy, and self-awareness. We also investigated the moderating function of brand-supported charitable activities on these relations. Additionally, we examined the causal relations of the three benefit variables.

From an academic standpoint, this study generally delves into the untapped potential of multiplayer online games to broaden players' perspectives—a subject that has yet to be extensively explored. On a practical level, fostering players' self-awareness has a direct positive effect on them whereas nurturing a deep connection with the gaming brand benefits companies. This strategy not only promotes the gaming community's well-being but also boosts the brand's presence among its audience, leading to a more cohesive global society.

Literature review

This section outlines three benefit variables that contribute to a win–win–win situation: FSDOP, brand intimacy, and self-awareness. It provides an overview of studies for each variable, highlighting key concepts and relevant findings. Subsequently, we examine three variables in multiplayer online gaming: social interactions in gaming, social interactions in gaming communities, and self-efficacy in gaming. We also introduce the role played by charitable activities in gaming, a practice that gaming brands have increasingly adopted. These studies build a theoretical foundation for logical connections among these variables and hypothesis development. The integration of these concepts, particularly in the gaming context, has remained largely unexplored.

Feelings of social connection with distant other players (FSDOP). Researchers have considerably examined the significance of fostering feelings of social connection with distant individuals (Cavanaugh et al. 2015). Such connections transcend mere social interactions and have profound implications for our overall well-being. Cavanaugh et al. (2015) highlighted the pivotal influence of such feelings on individuals' readiness to assist distant others. In today's interconnected world, many challenges and conflicts require global cooperation (Schwab 2021). If gamers can cultivate strong FSDOP, global players will have the opportunity to unite and collaborate in the resolution of various global issues such as climate change. However, despite its potential benefits, the cultivation of these feelings through multiplayer games remains underresearched.

Brand intimacy. Brand intimacy goes further than mere brand recognition and encompasses an emotional bond that fosters positive sentiment and attachment (Park et al. 2006). Scholars have highlighted several significant benefits of brand intimacy for a company, such as increased customer loyalty and retention (Keller and Aaker, 1992; Park et al. 2001), enhanced brand equity (Park et al. 2001; Park et al. 2006), better financial performance (Keller and Aaker 1992), and higher customer engagement and advocacy (Bairrada et al. 2018; Keller and Aaker 1992). Consumers experiencing brand intimacy tend to hold positive feelings about the brand and its values, choose it over competitors even at a premium, spread positive word-of-mouth, act as brand advocates, and remain loyal through challenges and setbacks (Bairrada et al. 2018). Furthermore, brand intimacy can also bridge social gaps (Almubarak et al. 2018).

Self-awareness. Self-awareness refers to one's conscious recognition and understanding of their character and uniqueness (Eurich 2018). A deep understanding of oneself promotes personal growth by helping identify individuals' values, goals, and strengths, thus enhancing their self-confidence (Showry and Manasa 2014). In multiplayer online games, although external factors can be interesting to examine, this study focuses solely on private self-awareness—that is, one's ability to recognize and understand their internal states such as thoughts, feelings, and motivations—as the fundamental pillar of personal development (Auzoult 2013; Grant et al. 2002; Ugur et al. 2015).

When people believe that they can achieve their goals, they are more likely to assess their strengths and weaknesses, improving their self-awareness, which can also be enhanced by reducing self-centeredness (Keng et al. 2011).

Social interactions in gaming. In the gaming context, social interactions refer to different forms of communication,

collaboration, and competition between players and other players or game characters in online games (Li et al. 2015; Merhi 2016). Social interactions in gaming can provide players with a unique platform to experience different characters, situations, and relationships in the virtual world, affecting their gaming experience, gaming motivation, and gaming satisfaction (Wu and Wang 2024).

Social interactions in gaming communities. Gaming communities take many forms—some created by game companies and some emerging organically, such as those on Steam or Twitch Chat. Players in organically formed communities enjoy greater autonomy, and their discussions often transcend the game itself to encompass broader aspects of gaming culture, personal experiences, and different topics of interest to the members. Participants in these communities can be complete strangers or individuals worldwide.

Self-efficacy in gaming. Bandura (1995) posited that self-efficacy is an individual's confidence in their ability to perform well in a specific situation. It is a dynamic factor that motivates people to pursue more challenging goals and persevere in overcoming obstacles to achieve such goals if they believe they are capable of doing so (Bandura 1995). In the context of playing video games, self-efficacy refers to an individual's belief in their ability to successfully accomplish gaming-related tasks, overcome challenges, and achieve desired outcomes (Pavlas 2010; Wu and Wang 2024). It can also influence their motivation, persistence, enjoyment, and performance in gaming activities (Wu and Wang 2024).

Gaming brands engaging in charitable activities. Studies have shown that everyone benefits when gaming brands engage in charitable initiatives (Kim et al. 2017). Brands improve their public image and promote a healthier gaming environment as evidenced by Xbox's partnership with the Special Olympics for inclusive gaming events (news.xbox.com) and Nintendo's support for various causes (www.nintendo.co.jp). Riot Games has also raised over $50 million through in-game fundraisers supporting causes such as education, citizenship, opportunity, and sustainability (riotgames.com). One notable example is their partnership with the Make-A-Wish Foundation to grant life-changing wishes for children with critical illnesses (riotgames.com). Players who participate in such prosocial activities increase their feelings of happiness and experience a stronger sense of belonging (Suryavanshi et al. 2024). Although research has confirmed the benefits of charitable gaming for brands (Li et al. 2023), this study conducts a deeper investigation by exploring how these activities influence the relations between various factors within a comprehensive model.

Hypotheses

The following four subsections present the study hypotheses. The first subsection addresses the causal relations of the three multiplayer online game variables. The second represents the study's core focus, discussing the relations between the three game variables and the three benefit variables. It also examines how charitable activities moderate the relation between the three game variables and brand intimacy as well as FSDOP. The third discusses the causal relations of the three benefit variables and examine how charitable activities moderate the relation between self-awareness and brand intimacy. Finally, the fourth examines how charitable activities influence brand intimacy and FSDOP.

Causal relations of the three multiplayer online game variables. In multiplayer games, teamwork allows players to discover strategies and enhance their gameplay, improving their performance (Fox et al. 2018). For gamers, the journey of overcoming in-game obstacles and celebrating collective victories is deeply satisfying (Morrison-Smith and Ruiz 2020). These interactions not only cultivate among players a sense of unity and inclusion but also drive them to continue playing and progressing (Yee 2006; Wu and Wang 2024). A player who receives affirmation from peers, such as praise and encouragement, can significantly elevate their self-assurance and perceived competence (Cohen and Sherman 2014). Ultimately, the nurturing aspect of the social environment of gaming is crucial in helping players improve their abilities, boost their confidence, and feel more empowered in their gaming pursuits. Therefore, we propose the following hypothesis:

H1: Players' social interactions in gaming positively affect their self-efficacy in gaming.

Multiplayer gaming often inspires players to share experiences and skills and discuss common interests (Gökçearslan and Alper 2023), which are precisely the functions that gaming communities excel at providing. Regular in-game interactions not only sharpen teamwork, negotiation and communication skills but also prepare players to engage in broader community forums, chats, or voice channels. When in-game challenges arise, players can obtain support from gaming communities, which typically focus on specific games or genres (Fox et al. 2018). In essence, the positive social dynamics that players experience in games serve as a foundation for comfortable and meaningful engagement within the expansive gaming community (Gandolfi et al. 2024). Hence, we propose the following hypothesis:

H2: Players' social interactions in gaming positively affect their social interactions in gaming communities.

One's confidence in their gaming skills often translates into greater initiative and social engagement in the gaming community (Von der Heiden et al. 2019). Players with strong self-efficacy in gaming not only aim to motivate and invigorate community participation but also adopt a collaborative spirit, rather than frustration, in addressing in-game challenges (Von der Heiden et al. 2019). Their belief in their capabilities makes them more inclined to resolve conflicts constructively, enhancing the social atmosphere for everyone. Overall, self-efficacy in gaming is a powerful force that allows players to become more proactive and more positive forces within the intricate social network of gaming communities. Therefore, we propose the following hypothesis:

H3: Players' self-efficacy in gaming positively affects their social interactions in gaming communities.

Relations between three multiplayer online game variables and three benefit variables and the moderating effects of charitable activities. Players with self-efficacy in gaming tend to define challenging objectives and dedicate themselves to accomplishing them (Wu and Wang 2024). People who experience success in one situation will increase their confidence to succeed in similar ones (Bandura 1995). Players who acquire self-efficacy in multiple areas are more confident, reinforcing their positive qualities and self-worth (Hoban and Hoban 2004). When an individual feels confident, they are not only secure about their strengths, but they also acknowledge their weaknesses more openly (Komarraju and Nadler 2013). Such willingness to embrace both positive and negative qualities improves their self-awareness. In addition, with self-efficacy, people believe they can learn a new skill and are more likely to invest effort in understanding their limitations (Bandura 1995). This process of identifying weaknesses and seeking improvement strategies ultimately leads one to understand themselves more comprehensively. Hence, we propose the following hypothesis:

H4: Players' self-efficacy in gaming positively affects their self-awareness.

Self-efficacy in gaming refers to a gamer's belief in their own abilities, strategies, and competencies (Wu and Wang 2024). This

self-assurance not only bolsters their emotional bond with the gaming brand but also leads them to attribute their success and positive experiences to the brand when it reinforces such feelings (Liu et al. 2019). Moreover, when gamers feel confident about their potential for success (self-efficacy) in the game, they are more inclined to feel accomplished and content with their performance. These rewarding experiences are then linked to the brand and thus intensify brand intimacy.

Players develop their sense of self-efficacy in gaming through their engagement with the game, building a connection with the gaming brand. However, a gaming brand that places significant emphasis on its charitable efforts may unintentionally shift players' attention away from honing their skills (Hamari and Koivisto 2015). Such a shift becomes more pronounced if charitable acts have no direct link to the gaming narrative or are not well-integrated into it. Players who value self-efficacy and the pursuit of skill enhancement view such unrelated interjections as irritation (Leary et al. 2007). For instance, a fighting game like *Street Fighter II* promoting charitable efforts unrelated to the in-game fight could potentially divert players who are primarily focused on skill mastery. In these scenarios, any perceived nonessential interruptions can weaken players' brand intimacy. Thus, we propose the following hypotheses:

H5a: Players' self-efficacy in gaming positively affects their brand intimacy.

H5b: This positive effect is weakened by the presence of charitable activities by gaming brands.

As discussed previously, one who achieves self-efficacy across various domains not only boosts their confidence but also fortifies their sense of self-worth and positive attributes (Hoban and Hoban 2004). These individuals typically expand their viewpoints and interests and embrace a curiosity for different cultures, values, and lifestyles, fostering a greater openness (Showry and Manasa 2014) that builds a more inclusive attitude toward people from diverse backgrounds. Recognizing and valuing their own positive qualities and achievements empowers individuals to confidently engage with new players (Bandura 1995), regardless of geographical distance. Therefore, we propose the following hypothesis:

H6: Players' self-efficacy in gaming positively affects their FSDOP.

Multiplayer games can develop into a realm of connection (Cole and Griffiths 2007). Social interactions within games instill in players a deeper layer of enjoyment (Cole and Griffiths 2007). Collaborative play or heated competition fosters shared experiences and emotions. Moreover, social interactions in gaming fuel positive reinforcement (Teng and Chen 2014). Players receive encouragement, praise, and recognition from their peers and therefore boost their motivation and enjoyment (Morrison-Smith and Ruiz 2020). Such positive experiences can create among players an emotional link with the game, fostering a stronger intimacy for the gaming brand that facilitated these connections (Mostafa and Kasamani 2020).

As discussed earlier, social interactions in gaming can provide players with a sense of belonging and emotional connections to the gaming brand. When a gaming brand engages in altruistic acts through charitable activities, which aim to help others, players deepen their emotional connection to the brand (Ellen et al. 2006). This positive emotional bond, created by the brand's altruistic behavior, further strengthens emotional ties formed through social interactions in gaming. Simply put, game companies that participate in charitable initiatives can enhance the positive impact of social interactions on brand intimacy. Hence, we propose the following hypotheses:

H7a: Players' social interactions in gaming positively affect their brand intimacy.

H7b: This positive effect is enhanced by the presence of charitable activities by gaming brands.

Online multiplayer games offer players a virtual space where they can transcend geographical boundaries. These platforms are especially advantageous for gamers in remote areas or those who find social interaction difficult (Morrison-Smith and Ruiz 2020). They forge a collective realm in which players pursue shared objectives, surmount challenges, or simply partake in camaraderie and connectivity (Chen 2009). Such mutual engagement nurtures team members' social connections and trust, fostering an emotional commitment to peers who may otherwise be difficult to connect with through traditional channels (Sloan and Oliver 2013). Therefore, online gaming has emerged as a useful tool for strengthening social bonds among players far and wide.

In addition, gaming companies that engage in charitable initiatives align themselves with the core principles of philanthropy, which are deeply rooted in the act of helping others and performing acts of kindness (Kim et al. 2017). Such generous behavior can cultivate among players a sense of unity, leading to the development of social connections (Meier and Stutzer 2008; Suryavanshi et al. 2024) and transcending physical distance. Moreover, a gaming brand's commitment to charitable causes demonstrates to players a shared dedication to altruism. Through interactive social experiences within the game, players experience a stronger sense of community belonging and reinforce social ties. Hence, we propose the following hypotheses:

H8a: Players' social interaction in gaming positively affects their FSDOP.

H8b: This positive effect is enhanced by the presence of charitable activities by gaming brands.

Gaming communities provide players with a vibrant platform to build connections, within which they exchange essential insights from game mechanics to personal experiences and strategic methods (Siitonen 2015). These discussions promote an environment of collaboration where members can both share and broaden their understanding (Hau and Kim 2011). These communities also foster a nurturing atmosphere filled with positivity and encouragement, inspiring players to reflect on their gameplay, identify areas for improvement, and actively seek progress (Papargyris and Poulymenakou 2005). Players who encounter obstacles can also find solace and assistance from peers within the community to overcome personal hurdles. In addition, gaming communities allow players to recognize their unique strengths and areas for improvement. Collectively, these elements enhance gamers' self-awareness. Furthermore, these social interactions constitute a powerful tool for reducing self-centeredness (Keng et al. 2011), allowing players to gain valuable insights into their own thoughts, feelings, and behaviors and consequently enhancing their self-awareness (Keng et al. 2011).

H9: Players' social interactions in gaming communities positively affect their self-awareness.

Active gaming communities are dynamic platforms for discussion and support, and social interactions within these communities can intensify players' sense of shared commitment (Nusair et al. 2013). When players proactively discuss the games they play and address topics such as the brands' strategies and playing experiences or shape a shared narrative about the games, their connection to the brand becomes stronger (Kim et al. 2019). Moreover, some gaming companies actively engage with these communities for direct communication, building a sense of recognition and value from the brand. Positive interactions with game developers can reinforce intimacy toward the brand (Casaló et al. 2007), especially when they are responsive to community feedback.

However, discussions within gaming communities typically focus on game-related topics such as gameplay experiences, strategies, and emotional responses (Gandolfi et al. 2024). When charitable activities dominate other aspects of social interaction, they may eclipse these

vital discussions, which are crucial for fostering brand intimacy among players. Also, when players perceive that they are pressured to engage in altruistic behaviors, they may feel embarrassment and discomfort (Leary et al. 2007). This pressure to conform to a philanthropic image can inadvertently diminish players' sense of intimacy with the gaming brand. For example, in a medieval fantasy role-playing game like *The Witcher 3: Wild Hunt*, the prominence of real-world charitable initiatives within gaming communities may create a disconnect between the game's historical fantasy setting and modern charitable efforts, potentially weakening its connection with the community. Thus, we propose the following hypotheses:

H10a: Players' social interactions in gaming communities positively affect their brand intimacy.

H10b: This positive effect is weakened by the presence of charitable activities by gaming brands.

Social interactions in gaming communities provide gamers worldwide with functions such as shared interest, collaboration and teamwork, a space for virtual interactions, and shared experiences (Cole and Griffiths 2007). The virtual world of online games offers them a wealth of shared experiences. Social interactions in gaming communities fulfill many of the same needs for social connection that people experience in face-to-face interactions (Cole and Griffiths 2007), providing a sense of belonging, fostering trust and cooperation, and allowing players to share experiences with others who understand their interests (O'Connor et al. 2015). Such a virtual world transcends physical boundaries, fostering meaningful social connections despite the distance.

In addition, similar to the discussion of H8b, we predict that the positive impact of players' social interactions in gaming communities on their FSDOP will be enhanced by the presence of charitable activities by gaming brands. Therefore, we propose the following hypotheses:

H11a: Players' social interactions in the gaming community positively affect their FSDOP.

H11b: This positive effect is enhanced by the presence of charitable activities by gaming brands.

Causal relations of the three benefit variables and the moderating effects of charitable activities. Self-aware individuals are cognizant of their strengths and weaknesses (Ashley and Reiter-Palmon 2012), the understanding of which is crucial as it marks the beginning of self-improvement (Ashley and Reiter-Palmon 2012). In multiplayer gaming, players with heightened self-awareness often choose roles that enhance team synergy or engage in strategic communication to boost coordination (Bosse and Gerritsen 2019). Their grasp of the team's social dynamics allows them to detect nuances in voice chats or gameplay, which can be pivotal during cooperative or competitive sessions. These insights deepen these players' understanding of interpersonal interactions within the game (Bosse and Gerritsen 2019; Showry and Manasa 2014). In addition, a multiplayer gaming brand that nurtures these moments of growth forges a stronger bond with its audience, particularly those who value the self-improvement journey. As a result, players develop a sense of intimacy toward this brand.

Self-aware players are also less prone to game addiction and are more attuned to their altruistic values and priorities (Froming et al. 1985). A gaming brand that supports altruistic causes can forge a more profound emotional bond with such gamers (Li et al. 2021). Players who engage in the brand's charitable endeavors are empowered to actively express their values, nurturing a collective sense of purpose and strengthening their affiliation with the brand. Thus, we propose the following hypotheses:

H12a: Players' self-awareness positively affects their brand intimacy.

H12b: This positive effect is enhanced by the presence of charitable activities by gaming brands.

Brand intimacy pertains to the feeling of connectedness with a brand and its community (Albert et al. 2009; Park et al. 2006). Players with the same degree of brand intimacy share values and experiences that can foster a sense of identity and belonging. Thus, gamers with brand intimacy often feel positive emotions toward the brand and its members, including trust, admiration, and enjoyment (Coelho et al. 2019). These emotions can also shape their interactions with others who are linked to the brand, making them more open and friendly. Additionally, discussing shared brand experiences can activate players' brain areas involved in reward and social connection (Alexandrov et al. 2013). This leads us to presume that brand intimacy for a gaming brand can help players feel more connected to other players even if they are far away. Therefore, we propose the following hypothesis:

H13: Players' brand intimacy positively affects their FSDOP.

Figure 1 illustrates the comprehensive model.

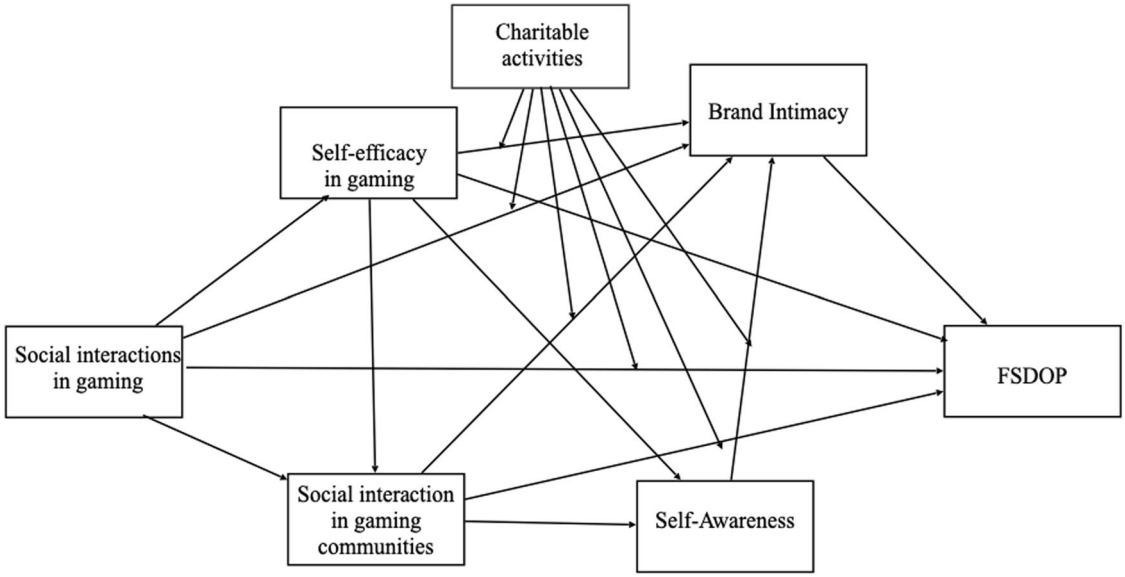


Fig. 1 The comprehensive model.

The effects of charitable activities on brand intimacy and FSDOP. Philanthropy is a trait shared by everyone and a value that most people hold. Studies have found that companies engaging in philanthropic pursuits will enhance customers' positive brand association (Kim et al. 2017), emotional connection (Green and Peloza 2011), and trust and goodwill (Park et al. 2016). Therefore, charitable activities led by gaming brands will increase brand intimacy among gamers.

In addition, a gaming brand that conducts effective charitable activities inspires this shared value among players (Майнка et al. 2023). Because players have common values, they feel more connected to one another and are more likely to trust and respect other players. Activating shared values among players can also motivate them to set common goals (Shteynberg and Galinsky 2011). When people work toward a common objective, they support and encourage each other and build strong relationships in the process. Moreover, helping others through activities brings happiness (Meier and Stutzer 2008), and many studies have observed the association between helping others and kindness, open-mindedness, and happiness (Umberson and Karas Montez 2010). Hence, we propose the following hypotheses:

H14a: The presence of charitable activities by gaming brands will enhance players' brand intimacy.

H14b: The presence of charitable activities by gaming brands will enhance players' FSDOP.

Methods

This study has three objectives: (1) to test the validity of its proposed model; (2) to examine how enhancing engagement in charitable activities affects brand intimacy and FSDOP; and (3) to determine each game variable's direct and indirect effects on self-awareness, brand intimacy, and FSDOP.

Experimental manipulation and measures. The experimental group read a scenario that included participating in a charitable activity. Referring to Wu and Wang's (2024) manipulation scenario, this study provided the following scenario for the experimental group:

Please imagine the following scenario:

According to the United Nations Food Security Report, an estimated 691 million to 783 million people worldwide will face hunger, and severe food shortages are accelerating the deterioration of children's growing environment.

Imagine you are playing a multiplayer online game. In addition to providing an online game community that allows players around the world to share game information and chat with each other, the gaming brand has launched the "Players Can Help Project," which uses limited game rewards to attract players to form teams to complete themed tasks. All proceeds from the program will be used to help solve the problem of hunger and malnutrition in food-deficient areas; at the same time, employees serving the gaming brand also engage in volunteer services to help local residents develop technical and vocational skills with practical actions to make them self-sufficient. Over the last decade, the project has amassed and contributed over $1 million, liberating numerous individuals from the grips of starvation, with approximately 21,000 children receiving the gift of a brighter future. Some beneficiaries have even matured to join local service squads, dedicating themselves to the betterment of their communities and enhancing healthcare provisions.

The control group received a scenario with no engagement in a charitable activity, and other factors were the same for both groups.

Procedure and measures. First, this study investigated players' game-playing status through such questions as "Have you ever played online games?" and "Have you ever played multiplayer online games?" and the frequency at which they played games. Afterward, they read the manipulation scenario. Then, they filled out a questionnaire with measures of brand intimacy, FSDOP, self-efficacy in gaming, social interaction in gaming, social interaction in gaming communities, and self-awareness. Finally, the participants' demographic information was obtained.

Measures of brand intimacy were adapted from those developed by Albert et al. (2009) and He et al. (2012), consisting of four items. Measures of the FSDOP were adapted from those developed by Lee, Robins (1995) and Cavanaugh et al. (2015), composed of three items. Measures of social interaction in gaming were adapted from those developed by Demetrovics et al. (2011), consisting of four items. Measures of social interaction in gaming communities were adapted from those developed by Chua (2002), with four items. Measures of self-efficacy in gaming were adapted from those developed by Allan (2010), consisting of five items. Measures of self-awareness were adapted from those developed by Auzoult (2013) and Grant et al. (2002), with four items. Table 1 shows the items for each construct, which were rated using a six-point scale (1 = strongly disagree, 2 = disagree, 3 = slightly disagree, 4 = slightly agree, 5 = agree, 6 = strongly agree). This measurement tool compels respondents to deliberate thoughtfully and select the response that accurately represents their favorable or unfavorable viewpoint (Wang and Wang, 2019).

Sampling, reliability, and validity. This research developed two distinct questionnaires using Google Forms and disseminated them randomly across prominent social media channels, including Facebook, and via email. The distribution was further facilitated by the services of tatoh.com.tw and drsurveydone.com, which disseminated these questionnaires randomly. Each participant was randomly given one of the two questionnaires. Data gathering took place from March 25 to May 24, 2024. Of the 998 responses collected, we analyzed only those with reported experiences in playing multiplayer online games, resulting in 844 valid respondents. A total of 403 respondents were assigned to the control group, while 441 were assigned to the experimental group. The demographic breakdown of these participants was as follows: 49.0% identifying as female and 51% identifying as male and 7.3% above 51 years old, 21.1% between 41 and 50 years old, 32.6% between 31 and 40 years old, 35.7% between 21 and 30 years old, and 3.2% being college students under 20 years old.

As shown in Table 1, Cronbach's alpha coefficients for the six dimensions ranged from 0.690 to 0.897. All measurement items in the structured components met the established standards, with factor loadings detailed in Table 1 as well. Table 2 provides the means and standard deviations of all variables.

The data-gathering approach of this investigation made it unfeasible to determine the exact rate of responses or evaluate any latent bias from nonrespondents. Common method variance (CMV) outcomes indicated that the data collection process was not compromised by CMV issues.

The study's composite reliability (CR) and average variance extracted (AVE) values met their respective thresholds, adhering to Fornell and Larcker's (1981) criteria and confirming convergent validity for all variables (Table 2). Moreover, the square roots of the AVE values (Table 3) confirmed discriminant validity within the structural equation modeling (SEM). An analysis of variance inflation factors revealed no multicollinearity issues.

Table 1 Items and Factor loadings using confirmatory factor analysis.

Items	Factor loading
Brand intimacy (Cronbach's α = 0.857)	
I have a warm and comfortable feeling about the gaming brand.	0.841
I feel emotionally close to the gaming brand.	0.868
I value the gaming brand greatly in my life.	0.830
I trust the gaming brand.	0.809
FSDOP (Cronbach's α = 0.841)	
I feel a real sense of belonging when I'm around players that I know from other countries.	0.864
I feel very close to other players from all over the world.	0.897
I feel connected to other players from around the world.	0.852
Social Interaction in gaming (Cronbach's α = 0.845)	
I met many different people through multiplayer online games.	0.823
I made new friends through multiplayer online games.	0.853
I think playing multiplayer online games is a good social experience.	0.865
When I play multiplayer online games, I always go out of my way to help other players when they need assistance.	0.762
Social interaction in gaming community (Cronbach's α = 0.868)	
I often interact with other players through gaming communities.	0.864
I feel comfortable to share my thoughts about the game with other players in the gaming community.	0.846
I am open to receive criticism and suggestions from other players in the gaming community.	0.821
I'm glad to participate in gaming community discussions.	0.857
Self-efficacy (Cronbach's α = 0.841)	
If I had to play an online game that I had never played before, I still believed I can win.	0.771
If I see a friend succeeding at an online game, it gives me confidence that I can be succeed as well.	0.797
I can quickly and easily learn the buttons and controls for a new game.	0.787
No matter how challenging the online game is, I can beat it if I try hard enough.	0.827
If my friends encourage me, I can usually play better.	0.725
Self-awareness (Cronbach's α = 0.794)	
I am usually conscious of my inner feelings.	0.822
I am often reflective about my life.	0.690
I usually aware of my innermost thoughts.	0.817
I usually have a very clear idea about why I have behaved in a certain way.	0.815

Table 2 The mean, standard deviation, CR and AVE.

	M	SD	CR	AVE
Brand intimacy	4.4276	0.87716	0.904	0.701
FSDOP	4.1499	1.01132	0.904	0.759
Social interaction in gaming	4.6335	0.91470	0.896	0.683
Social interaction in gaming communities	4.3292	1.01285	0.910	0.718
Self-efficacy	4.3559	0.86738	0.887	0.612
Self-awareness	4.6904	0.72520	0.867	0.621

Model fit. This study employed SEM using smartPLS 4 to test the proposed hypotheses. Our model incorporated a binary moderator variable alongside other variables measured on a six-point scale. Also, because the data originated from two distinct groups with differences in means and standard deviations (Table 4), we reported the unstandardized results for better interpretability. For model fit, the standardized root mean square residual (SRMR) was 0.058, which was comfortably below the acceptable threshold of 0.08. Additionally, the normed fit index (NFI) was 0.825; while this was below the commonly recommended value of 0.9, it did not detract from the overall model fit. Further validation through bootstrap-based testing for precise model fit yielded d_ULS = 1.465 and d_G = 0.385, which were below their respective critical upper bounds at the 99% (1.594) and 95% confidence levels (0.884), respectively. Collectively, these metrics robustly verified that our research model displayed a satisfactory fit.

Hypothesis analysis. Table 5 shows the path analysis results. The coefficient for social interaction in gaming and self-efficacy in gaming was 0.573 ($t = 17.098$, $p < 0.001$), which supports H1. The coefficient for social interaction in gaming and social interaction in gaming communities was 0.666 ($t = 16.537$, $p < 0.001$), supporting H2. The coefficient for self-efficacy in gaming and social interaction in gaming communities was 0.303 ($t = 7.294$, $p < 0.001$), supporting H3.

The coefficient for self-efficacy in gaming and self-awareness was 0.288 ($t = 7.351$, $p < 0.001$), supporting H4. The coefficient for self-efficacy in gaming and brand intimacy was 0.315 ($t = 4.736$, $p < 0.001$), supporting H5a. The moderating effect of charitable activities on this association yielded a coefficient of -0.277 ($t = 2.812$, $p = 0.005$), which supports H5b. The coefficient for self-efficacy in gaming and FSDOP was 0.160 ($t = 3.355$, $p = 0.001$), which supports H6.

The coefficient for social interaction in gaming and brand intimacy was 0.211 ($t = 3.200$, $p = 0.001$), supporting H7a. However, the moderating effect of charitable activities on the association between social interaction in gaming and brand intimacy yielded a coefficient of 0.141 ($t = 1.553$, $p = 0.121$), rejecting H7b. The coefficient for social interaction in gaming and FSDOP was 0.167 ($t = 2.418$, $p = 0.016$), which supports H8a. Meanwhile, the moderating effect of charitable activities on this link yielded a coefficient of 0.096 ($t = 0.877$, $t = 1.041$, $p = 0.298$), rejecting H8b.

The coefficient for social interactions in gaming communities and self-awareness was 0.135 ($t = 3.853$, $p < 0.001$), supporting H9. The coefficient for social interaction in gaming communities and brand intimacy was 0.190 ($t = 3.233$, $p = 0.001$), which supports H10a. The moderating effect of charitable activities on this association yielded a coefficient of -0.209 ($t = 2.322$, $p = 0.020$), which supports H10b. The coefficient for social interaction in gaming communities and FSDOP was 0.282

Table 3 The result of item Inter-variable correlations and square roots of AVE.

item	Brand intimacy	FSDOP	Social interaction in gaming	Social interaction in gaming communities	Self-efficacy	Self-awareness
Brand intimacy	**0.837**					
FSDOP	0.607	**0.871**				
Social interaction in gaming	0.548	0.624	**0.826**			
Social interaction in gaming communities	0.511	0.623	0.757	**0.847**		
Self-efficacy	0.497	0.554	0.591	0.625	**0.782**	
Self-awareness	0.418	0.373	0.376	0.387	0.443	**0.788**

Bold: the results of the square roots of AVE.

Table 4 The means, standard deviations in the control group and experimental group.

	Control group		Experimental group	
Construction	**M**	**SD**	**M**	**SD**
FSDOP	4.0310	1.0790	4.2562	0.9352
Social interactions in gaming	4.5653	0.9583	4.6944	0.8705
Social interactions in gaming communities	4.2299	1.0485	4.4180	0.9725
Self-efficacy in gaming	4.2789	0.8914	4.4247	0.8403
Self-awareness	4.6376	0.7568	4.7376	0.6931
Brand Intimacy	4.2463	0.90009	4.5971	0.82069

($t = 4.255$, $p < 0.001$), supporting H11a. The moderating effect of charitable activities on this link yielded a coefficient of -0.090 ($t = 1.074$, $p = 0.283$), which rejects H11b.

The coefficient for self-awareness and brand intimacy was 0.115 ($t = 2.087$, $p = 0.037$), supporting H12a. The moderating effect of charitable activities on this association yielded a coefficient of 0.198 ($t = 2.328$, $p = 0.020$), supporting H12b. The coefficient for brand intimacy and FSDOP was 0.349 ($t = 7.883$, $p < 0.001$), which supports H13.

In addition, the results in Table 4 indicate that the experimental group had significantly greater brand intimacy than the control group (Mtreatment = 4.5971 vs. Mcontrol = 4.2463, $t = 5.819$, df = 842, $p < 0.001$). These results support H14a. Table 4 also shows that the experimental group had significantly greater FSDOP than the control group (Mtreatment = 4.2562 vs. Mcontrol = 4.0310, $t = 3.560$, df = 842, $p < 0.001$). These results support H14b. Overall, while "the presence of charitable activities" is beneficial for brand intimacy, it also enhances FSDOP.

Discussion

The moderating effect of charitable activities presents an intriguing observation. The current findings reveal that charitable activities can weaken the association between self-efficacy and brand intimacy while enhancing the effect of self-awareness on brand intimacy as described by H5b and H12b. One example is Riot Games' support for mental health awareness initiatives. When players are still developing their self-efficacy in gaming, introducing charitable activities can be distracting, diminishing players' sense of brand intimacy. In contrast, players who have established stronger self-awareness through gaming self-efficacy and social interactions within gaming communities experience enhanced brand intimacy brought about by charitable activities. This is because players with heightened self-awareness are less likely to be purely self-centered or immersive when engaging with a game (Keng et al. 2011; Ugur et al. 2015) and instead approach their gaming experience more objectively, recognizing it as a virtual activity. When a game company introduces charitable initiatives, these players tend to express a higher appreciation for the altruistic values of such efforts. This rational assessment then fosters a deeper emotional connection with the brand, strengthening brand intimacy.

Relations between variables that are not moderated by the presence of charitable activities, as in H7b, H8b, and H11b, may be attributed to the strong direct effects between such variables, which render the moderating influence of charitable activities insignificant.

Conclusion

Theoretical implications. Multiplayer online games provide global players with an effective platform to communicate, connect, and build relationships. By exploring the largely untapped potential of multiplayer online games, this study investigated how such virtual environments can produce a win–win–win situation for players, gaming brands, and the broader global gaming community, allowing them to collaboratively address global challenges. Other key contributions of this study are as follows.

First, we crafted a comprehensive model that meticulously structures the behaviors and psychological patterns of gamers participating in multiplayer online games. This model begins with an exploration of players' social interactions in gaming and progresses to an examination of their influence on players' gaming self-efficacy, social interactions in gaming communities, self-awareness, brand intimacy, and FSDOP. Subsequently, this study examined the intricate relations among these variables and introduced the moderating role played by charitable activities conducted by gaming brands. This comprehensive framework allows for the creation of strategies that specifically reinforce gamers' self-awareness, brand intimacy, and FSDOP. Our model represents a pioneering effort that offers novel insights in gaming behavior research.

Second, this study confirmed that players' social interactions in gaming are crucial in strengthening their bond with gaming brands as well as in nurturing global social connections irrespective of philanthropic endeavors. In terms of fostering FSDOP and brand intimacy, the comprehensive effects of players' social interactions in gaming surpassed those of all other examined variables. Specifically, the total effect of social interactions in gaming on FSDOP was 0.699, while the total effect on brand intimacy was 0.583, highlighting the dominant and accumulative impact of social interactions in gaming. In addition, social interactions in gaming communities focusing on multiplayer online games played some significant roles including bolstering players' brand intimacy and FSDOP.

Third, in cultivating a robust connection between players and their FSDOP, influence comes not just from social interaction in

Table 5 The path analyses.

			Estimate	*T*-value	*P*-value
Social interactions in gaming	-->	Self-efficacy in gaming	0.573	17.098	***
Social interactions in gaming	-->	Social interactions in gaming communities	0.666	16.537	****
Social interactions in gaming	-->	Brand intimacy	0.211	3.200	0.001*
Social interactions in gaming	-->	FSDOP	0.167	2.418	0.016*
Self-efficacy in gaming	-->	Social interactions in gaming communities	0.303	7.294	***
Self-efficacy in gaming	-->	Self-awareness	0.288	7.351	***
Self-efficacy in gaming	-->	Brand intimacy	0.315	4.736	***
Self-efficacy in gaming	-->	FSDOP	0.160	3.355	0.001*
Social interactions in gaming communities	-->	Self-awareness	0.135	3.853	***
Social interactions in gaming communities	-->	Brand intimacy	0.190	3.233	0.001*
Social interactions in gaming communities	-->	FSDOP	0.282	4.255	***
Self-awareness	-->	Brand intimacy	0.115	2.087	0.037*
Brand intimacy	-->	FSDOP	0.349	7.883	***
Charitable activities X Social interactions in gaming	-->	Brand intimacy	0.141	1.553	0.121
Charitable activities X Social interactions in gaming	-->	FSDOP	0.096	1.041	0.298
Charitable activities X Social interactions in gaming	-->	Brand intimacy	−0.277	2.812	0.005*
Charitable activities X Self-efficacy	-->	Brand intimacy	−0.209	2.322	0.020*
Charitable activities X Social interactions in gaming communities	-->	FSDOP	−0.090	1.074	0.283
Charitable activities X Social interactions in gaming communities	-->	Brand intimacy	0.198	2.328	0.020*
Charitable activities X Self-awareness					

*<0.05, **<0.01, ***<0.001.

gaming and gaming communities; players' brand intimacy and self-efficacy in gaming are also pivotal. The total effect of brand intimacy on FSDOP was 0.349, while that of gaming self-efficacy on FSDOP was 0.389. Both were notably higher than or nearly equivalent to the total effect of social interaction in gaming communities on FSDOP, which was 0.354. These findings further deepen our understanding of the factors that foster social connections in the gaming sphere.

Fourth, as anticipated, engagement in charitable activities can enhance players' brand intimacy and FSDOP. Nevertheless, the moderating effects are worth discussing. For FSDOP, charitable activities did not moderate the influence of social interactions in gaming and social interactions in gaming communities. Regarding brand intimacy, while such engagement can amplify the positive impact of players' self-awareness on their brand intimacy, it concurrently diminishes the impact of players' gaming self-efficacy and social interactions in gaming communities on their brand intimacy. One can expect that most gamers with strong gaming self-efficacy will seek to further enhance their skills. When players take part in gaming communities, they tend to direct their attention toward game-specific topics such as gameplay, strategy, and emotional responses. In such contexts, they may find charitable activities a distraction, potentially leading to negative interactions. These findings indicate that while charitable engagement can be beneficial, it must be approached with an understanding of its varying effects on different aspects of player experience and community dynamics.

Finally, our research findings show that by engaging in philanthropy, multiplayer game companies can create a win–win–win situation for players, game companies, and global society, enhancing players' social interactions, gaming self-efficacy, and self-awareness.

Practical and social implications. Multiplayer games have a distinctive potential to improve social interactions beyond the physical world, cultivating a global community and fostering connections among players worldwide. To maximize this potential, game developers are encouraged to create engaging social features such as robust in-game communication tools and community-centric events such as proximity chat and in-game festivals, which can inspire players to unite and collaborate. Specifically, the inclusion of players from different countries is highly advocated. By implementing these strategies, multiplayer games can transform into influential platforms for fostering meaningful relationships and a deep sense of belonging among players internationally, complementing their gaming experience and brand allegiance.

Moreover, the integration of impactful charitable activities into games can highlight philanthropy among players, which may strengthen their brand intimacy and foster FSDOP. Our findings show that altruistic initiatives not only benefit others but also enhance brand value. Meanwhile, the results also reveal that charitable efforts must be carefully managed as they may weaken the impact of players' self-efficacy and social interactions within gaming communities on brand intimacy. To navigate this potential conflict, charitable elements must be harmonized with core game design. While different game genres may require tailored approaches, there is still an opportunity to incorporate charitable initiatives. For example, developers of war-themed

games could introduce programs that support war-affected civilians. Players' in-game achievements, such as reaching specific milestones, could prompt the company to donate a portion of its revenue to assist such individuals. These initiatives may seem counterintuitive to the nature of war games, but they can yield significant benefits by enhancing the company's brand image, promoting social responsibility, and enriching the game's thematic depth. Conversely, in genres where charitable actions can more naturally align with the gaming experience, they can further amplify positive outcomes without focusing on personal gain. For instance, city-building simulation games could be designed to encourage players to consider global sustainability, shifting the focus from short-term expansion to fostering a win–win scenario for both individual players and the broader community.

Next, this study highlighted the importance of enhancing players' self-awareness during charitable activities. Developers can design in-game experiences that encourage players to reflect on and assess their gaming abilities. Cultivating gaming self-efficacy and engaging in such evaluations can gradually enhance players' self-awareness (Grant et al. 2002). Deeper self-awareness among gamers can then increase their appreciation for charitable efforts by enhancing their self-signal utilities (Ugur et al. 2015; Savary et al. 2015). These activities not only boost players' self-awareness but also strengthen their positive engagement with the gaming brand and social connection with gamers worldwide.

Consistent with the present findings, the cultivation of players' gaming self-efficacy can lead to substantial improvements in their self-awareness, brand intimacy, and FSDOP. Gaming brands should therefore work toward improving players' self-efficacy through well-defined and achievable objectives. Players must experience a sense of progression and accomplishment even with smaller goals (Wang et al. 2017).

In conclusion, effective strategies in multiplayer games can generate mutually beneficial scenarios for players, game companies, and the global community. Specifically, by nurturing connections among distant players, gaming brands may reinforce social bonds, increase empathy and understanding worldwide, and foster a sense of community. If more gaming companies embraced these practices, they could significantly contribute to the resolution of global issues.

Research limitations and future research directions. This study has several limitations that open opportunities for future research. First, it presented the unstandardized results with the moderator. The model in this study incorporated a binary moderator variable alongside other variables measured on a six-point scale. Notably, the data were extracted from two distinct groups with varying means and standard deviations (Table 4). This necessitated the reporting of unstandardized results for clearer interpretation. Despite achieving a good fit with the unstandardized results, we encountered a trade-off: standardization could mislead the interpretation. To address this, we considered two separate models, but doing so would hinder the comparison of relations between variables with and without charitable activities. Consequently, we decided to present the unstandardized results alongside the moderator. Future research efforts can explore strategies to overcome these challenges.

Second, in this study, measures of social interactions in gaming and in gaming communities gauged players' overall behaviors instead of focusing on individual gaming franchises. This approach was rooted in the observation that gamers' activities and community engagement often surpass the boundaries of individual game publishers and their associated groups. As a result, our research cast a wider net and was not limited to particular gaming brands. Future studies could examine the impact of exclusively focusing on a single gaming brand and its community, which might produce unique findings.

Third, this study investigated the nature of charitable activities. Importantly, different types of charitable endeavors may elicit varied reactions. People's diverse attitudes toward these activities can influence their responses to other variables.

Fourth, the current results indicate that charitable activities may attenuate the influence of players' self-efficacy and social interactions in gaming communities on brand intimacy. A promising research opportunity involves an investigation of how to incorporate charitable elements into game design in a manner that mitigates this potential conflict or, better yet, synergistically aligns both charitable impact and player engagement.

Fifth, this study did not fully account for the reciprocal relation between social interactions and self-awareness. Individuals with strong self-awareness may positively shape their social interactions, a dynamic that deserves further investigation. Future studies could use longitudinal data or cross-lagged panel models to examine this bidirectional effect more rigorously.

Additionally, this study specifically focused on private self-awareness, which is readily measurable through self-report methods because of its emphasis on gamers' conscious internal experiences. However, because self-awareness is a multifaceted construct, future studies should explore its various dimensions in more depth.

Finally, another limitation was the lack of categorization by participants' nationality and cultural background or game type in the research design. Different categories might yield different findings. Hence, future research can validate the proposed model by classifying respondents by nationality or cultural background and differentiating between various game types. Such an approach may lead to new discoveries.

Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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Author contributions

Couchen Wu and Chih-Hsuan Chang jointly conceived the research idea. Chih-Hsuan Chang was responsible for data collection, while both authors collaborated on developing the theoretical framework, conducting data analysis, and verifying the analytical methods. They also jointly discussed the results and contributed to the final manuscript, with Couchen Wu overseeing the final review and quality check.

Competing interests

The authors declare no competing interests.

Ethical approval

This study was conducted in compliance with all relevant institutional, local, and national regulations regarding research involving human participants. In line with the Taiwan Ministry of Health and Welfare guidelines (Article 1), certain studies are exempt from ethics committee review when specific conditions are met: participants are adults who do not belong to vulnerable populations, and the study uses non-interactive, non-invasive, non-identifiable data collection methods in public settings. This research specifically involved collecting online opinions from adult gamers about their gaming experiences. The data collection was carried out in public venues, using anonymous, non-interactive, and non-invasive techniques that ensured no personal identification of participants from the collected data. As a result, ethics committee approval was not required.

Informed consent

All participants were informed about the purpose of the study and that their data would be used solely for academic research purposes. Participation was entirely voluntary, and participants were free to discontinue the questionnaire at any time without facing any repercussions. The data collection process was anonymous and designed to ensure that no personally identifiable information was gathered. As the data was collected and analyzed in a manner that prevented the identification of specific individuals, there were no privacy concerns.

Additional information

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